

# RAOUL ISLAND, New Zealand

WMO: 939940

Lat: 29.245S

Lon: 177.929W

Elev: 162

StdP: 14.61

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 8             | 52.3       | 53.8 | 42.5                          | 40.3 | 58.6 | 44.1 | 42.8 | 58.7 | 21.2                  | 63.3 | 19.1 | 63.4 | 4.2                   | 220  | 0.322 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 7.8                    | 80.6            | 74.4 | 79.1 | 73.1 | 77.9 | 72.0 | 76.2                | 79.0 | 75.0 | 77.6 | 73.8 | 76.6 | 6.6                  | 100  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 75.3                            | 133.3 | 78.0 | 74.0 | 127.7 | 77.0 | 72.9 | 122.8 | 76.0 | 39.8          | 79.1 | 38.6 | 77.6 | 37.6 | 76.7 | 82.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 20.3          | 17.9 | 16.0 | DB                         | 49.7 | 83.6               | 1.4 | 1.9                                                | 48.7 | 84.9       | 47.8 | 86.0       | 47.0 | 87.1       | 46.0 | 88.4 | (4) |
| (5)               |      |      | WB                         | 46.9 | 77.5               | 1.2 | 1.5                                                | 46.0 | 78.6       | 45.3 | 79.5       | 44.6 | 80.3       | 43.7 | 81.4 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 66.9   | 72.4  | 73.5  | 72.8  | 70.0  | 66.5  | 63.4  | 61.8  | 60.9  | 62.1  | 63.7  | 66.4  | 69.7  | (6)  |  |
|                                                                        | DBStd       | 5.13   | 2.69  | 2.28  | 2.37  | 2.79  | 2.58  | 2.93  | 2.78  | 2.57  | 2.47  | 2.43  | 2.74  | 2.61  | (7)  |  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |  |
|                                                                        | HDD65       | 477    | 0     | 0     | 0     | 1     | 14    | 64    | 106   | 130   | 91    | 54    | 16    | 1     | (9)  |  |
|                                                                        | CDD50       | 6169   | 696   | 659   | 707   | 599   | 511   | 402   | 364   | 337   | 364   | 426   | 492   | 612   | (10) |  |
|                                                                        | CDD65       | 1173   | 231   | 239   | 242   | 150   | 60    | 17    | 6     | 2     | 5     | 15    | 58    | 148   | (11) |  |
|                                                                        | CDH74       | 2154   | 568   | 689   | 546   | 155   | 6     | 0     | 0     | 0     | 0     | 0     | 16    | 174   | (12) |  |
|                                                                        | CDH80       | 66     | 26    | 25    | 14    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 7.4    | 8.0   | 8.2   | 8.1   | 7.7   | 7.5   | 7.5   | 7.5   | 7.1   | 7.1   | 6.7   | 6.5   | 6.8   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 57.8   | 4.5   | 5.5   | 5.7   | 4.8   | 4.7   | 5.7   | 5.6   | 5.1   | 4.4   | 3.2   | 3.6   | 3.9   | (15) |  |
|                                                                        | PrecMax     | 84.4   | 18.3  | 14.4  | 15.9  | 21.2  | 10.0  | 14.8  | 12.6  | 13.4  | 10.2  | 9.2   | 12.5  | 14.8  | (16) |  |
|                                                                        | PrecMin     | 34.1   | 0.5   | 0.4   | 1.1   | 0.8   | 1.1   | 1.8   | 1.9   | 0.4   | 1.5   | 0.6   | 0.1   | 0.3   | (17) |  |
|                                                                        | PrecStd     | 12.8   | 3.8   | 3.6   | 3.7   | 3.4   | 2.2   | 2.8   | 2.3   | 2.3   | 2.2   | 1.8   | 2.6   | 3.0   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 82.7  | 82.5  | 81.8  | 79.2  | 74.8  | 72.3  | 70.2  | 69.2  | 70.6  | 72.4  | 76.0  | 79.4  | (19) |  |
|                                                                        |             | MCWB   | 76.6  | 76.0  | 76.2  | 73.6  | 68.2  | 68.0  | 64.9  | 64.5  | 64.9  | 66.2  | 69.9  | 73.0  | (20) |  |
|                                                                        | 2%          | DB     | 80.2  | 80.6  | 79.6  | 77.2  | 73.1  | 70.5  | 68.8  | 67.7  | 69.0  | 70.9  | 74.1  | 77.4  | (21) |  |
|                                                                        |             | MCWB   | 74.2  | 73.8  | 74.3  | 71.6  | 66.9  | 66.2  | 63.6  | 62.5  | 63.3  | 64.9  | 68.1  | 71.2  | (22) |  |
|                                                                        | 5%          | DB     | 78.3  | 79.1  | 78.2  | 75.7  | 71.9  | 69.3  | 67.5  | 66.6  | 67.8  | 69.7  | 72.8  | 76.0  | (23) |  |
|                                                                        |             | MCWB   | 72.2  | 72.6  | 72.8  | 70.3  | 66.0  | 64.9  | 62.6  | 61.2  | 62.5  | 63.7  | 66.9  | 70.0  | (24) |  |
|                                                                        | 10%         | DB     | 76.9  | 77.7  | 76.9  | 74.5  | 70.8  | 68.3  | 66.4  | 65.6  | 66.8  | 68.5  | 71.5  | 74.6  | (25) |  |
|                                                                        |             | MCWB   | 71.1  | 71.8  | 71.4  | 69.3  | 65.2  | 63.7  | 61.6  | 60.3  | 61.6  | 62.8  | 65.9  | 68.9  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 77.8  | 77.2  | 77.4  | 75.0  | 70.7  | 69.9  | 67.2  | 66.6  | 67.2  | 68.3  | 72.4  | 74.6  | (27) |  |
|                                                                        |             | MCDB   | 80.7  | 81.0  | 80.1  | 77.4  | 73.1  | 71.1  | 69.0  | 68.1  | 68.7  | 70.6  | 74.2  | 77.5  | (28) |  |
|                                                                        | 2%          | WB     | 76.0  | 75.7  | 75.9  | 73.4  | 69.1  | 68.0  | 65.7  | 64.7  | 65.6  | 67.0  | 70.3  | 72.9  | (29) |  |
|                                                                        |             | MCDB   | 78.7  | 78.5  | 78.5  | 75.7  | 71.5  | 69.7  | 67.6  | 66.6  | 67.3  | 69.3  | 72.4  | 75.7  | (30) |  |
|                                                                        | 5%          | WB     | 74.3  | 74.6  | 74.6  | 72.0  | 68.0  | 66.5  | 64.1  | 63.2  | 64.4  | 65.7  | 68.8  | 71.7  | (31) |  |
|                                                                        |             | MCDB   | 77.2  | 77.3  | 77.2  | 74.7  | 70.6  | 68.5  | 66.5  | 65.5  | 66.6  | 68.2  | 71.3  | 74.4  | (32) |  |
|                                                                        | 10%         | WB     | 72.7  | 73.6  | 73.2  | 70.8  | 66.7  | 65.0  | 62.7  | 61.8  | 63.2  | 64.4  | 67.4  | 70.3  | (33) |  |
|                                                                        |             | MCDB   | 75.8  | 76.5  | 75.8  | 73.7  | 69.8  | 67.5  | 65.7  | 64.6  | 65.9  | 67.3  | 70.3  | 73.6  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.1   | 7.8   | 7.5   | 7.6   | 7.5   | 7.4   | 7.6   | 8.4   | 8.3   | 8.8   | 8.8   | 8.7   | (35) |  |
|                                                                        |             | MCDBR  | 9.6   | 9.7   | 8.7   | 8.7   | 8.2   | 7.9   | 8.2   | 9.0   | 9.2   | 9.8   | 9.8   | 9.9   | (36) |  |
|                                                                        |             | MCWBR  | 5.3   | 5.3   | 4.8   | 5.2   | 5.2   | 5.9   | 5.9   | 6.0   | 6.0   | 6.1   | 5.8   | 5.6   | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 7.9   | 7.8   | 7.3   | 7.3   | 6.4   | 6.3   | 6.7   | 7.1   | 7.9   | 7.5   | 7.5   | 7.7   | (38) |  |
|                                                                        |             | MCWBR  | 4.9   | 5.0   | 4.6   | 5.3   | 5.3   | 6.2   | 6.0   | 6.1   | 5.8   | 6.0   | 5.4   | 5.2   | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.362  | 0.362 | 0.356 | 0.353 | 0.337 | 0.333 | 0.328 | 0.329 | 0.345 | 0.342 | 0.342 | 0.342 | 0.354 | (40) |  |
|                                                                        | taud        | 2.606  | 2.627 | 2.642 | 2.632 | 2.651 | 2.632 | 2.637 | 2.628 | 2.561 | 2.566 | 2.599 | 2.605 |       | (41) |  |
|                                                                        | Ebn at Noon | 311    | 305   | 297   | 282   | 272   | 266   | 273   | 286   | 296   | 308   | 315   | 314   |       | (42) |  |
|                                                                        | Edh at Noon | 33     | 31    | 29    | 27    | 24    | 23    | 24    | 26    | 31    | 33    | 33    | 33    |       | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 2206   | 2003  | 1646  | 1316  | 1011  | 866   | 959   | 1209  | 1577  | 1961  | 2175  | 2290  |       | (44) |  |
|                                                                        | RadStd      | 152    | 126   | 106   | 104   | 90    | 43    | 64    | 55    | 118   | 95    | 136   | 147   |       | (45) |  |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page